	$= 6.20 \times 10$ m s
7(a)	The work function energy of zinc is lower than the energy of the ultraviolet photons, but greater than that of red light photons. In order for photoelectric effect to take place, the energy and frequency of the incident photon must be higher than that of the metal's work function and threshold frequency respectively. When illuminated with red light, the energy of the photon is lower than the work function of the metal. Hence, no photoelectrons are ejected with red light.
7(b)	When the *β*-particles (electrons) are decelerated within the aluminium container, the kinetic energy lost by the electrons is partially converted into the continuous braking radiation of the X-ray, and partially into the characteristic spectrum of aluminium as a result of knocking out energy shell electrons electrons from aluminium and subsequent de-excitation of higher shell electrons into the vacated shells, resulting in emission of characteristic X-ray radiation. As X-rays are penetrating radiation, they are able to escape from the container walls.

7(c)	The atom is mostly empty space with a nucleus at its centre which is small in size, large in mass and positively charged. Hence, most of the α-particles incident pass through the empty space in the atom undeflected. Only a small number of positively-charged α-particles that come close to the positively-charged nucleus are deflected since like charges repel. In order to be deflected by more than 90°, the α-particles have to come within a small distance of the nucleus, in which the probability is very low due to its extreme small size.
8(a)(i)	Two bodies are in thermal equilibrium means they are at the same temperature.